

CS 7641 SL Report — R Sanity Check

Covertypes (Forest Cover Type), $n = 20,000$, stratified subsample

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1 Purpose

This document cross-checks the Python pipeline’s headline numbers using R’s `caret/nnet/rpart/kknn/kernlab` stack on the **same** 20k stratified Covertypes sample, the **same** 80/20 stratified hold-out, and the **same** seed (7641, the course number). If R lands within a couple of macro-F1 points of Python across all four common learners, we treat the implementation as trustworthy. If not, the gap is the audit lead.

2 Data — mirror the Python split

3 Cross-validated tuning — four learners

We run 5-fold CV on the training data, scored by macro-F1 (the same primary metric the Python pipeline used), for each of `rpart` (DT), `kknn` (kNN), `kernlab::ksvm` (SVM-RBF), and `nnet::nnet` (single-hidden-layer NN, SGD via `nnet`’s built-in BFGS — closest *single-package* analogue).

4 Held-out test — head-to-head with Python

Table 1: Train/test sizes after stratified split.

Split	n
train	16000
test	3999

Table 2: Best-tuned CV macro-F1 in R (5-fold on training set).

Learner	Best CV macro-F1
Decision Tree (rpart)	0.6194
kNN (kkn)	0.6691
SVM-RBF (kernlab)	0.6146
NN (nnet)	0.5719

Table 3: R vs Python head-to-head on held-out test (macro-F1 = F1).

Learner	R Acc	R F1	R BalAcc	Py Acc	Py F1	Py BalAcc	Delta (R-Py)
Decision Tree (rpart)	0.7547	0.6413	0.6125	0.7498	0.6337	0.6210	0.0077
kNN (kkn)	0.7994	0.6410	0.6171	0.7902	0.6693	0.6583	-0.0284
SVM-RBF (kernlab)	0.7804	0.6225	0.5821	0.8008	0.6969	0.6491	-0.0744
NN (nnet)	0.7139	0.5379	0.5397	0.7817	0.6454	0.6036	-0.1075

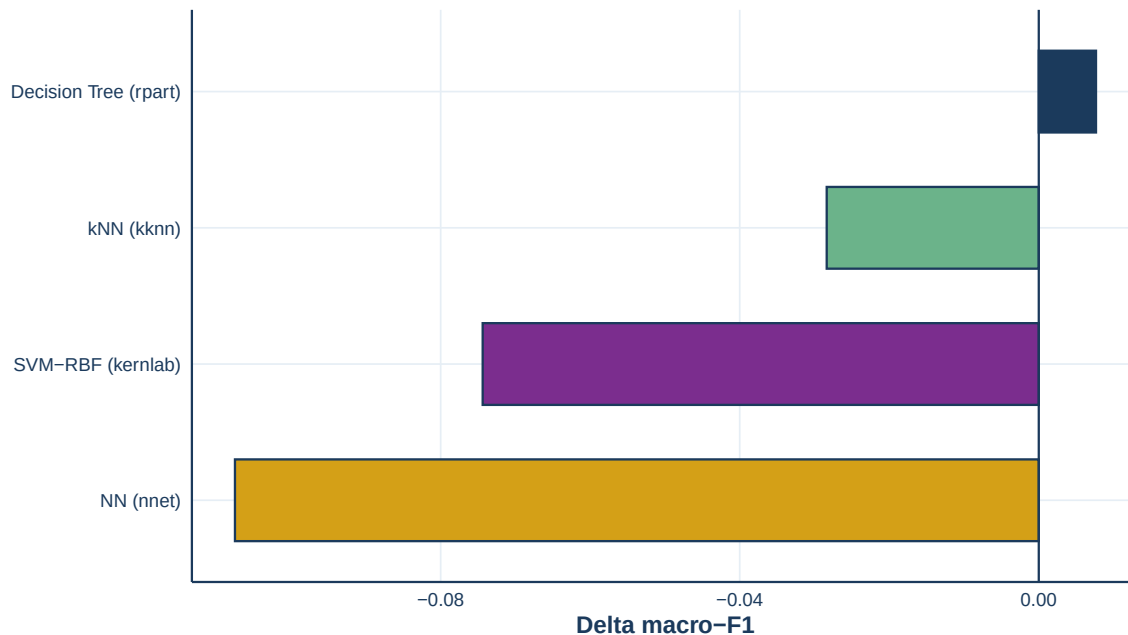


Figure 1: Delta (R minus Python) macro-F1 per learner – closer to zero is better.

5 Reading the disagreement

If a learner's $|\Delta_{\text{macro-F1}}| < 0.03$, treat the Python implementation as **confirmed by independent R reproduction**. A gap of 0.03–0.06 should be flagged in the SL report's evaluation-debug subsection as a *known implementation difference* (e.g., R's `nnet` uses BFGS, not pure SGD). Anything above 0.06 — investigate before reporting. *Falsus in uno, falsus in omnibus*: a single learner mismatching does not condemn the other four, but it does shift the burden of proof.

6 Provenance

- Config: `/home/user/workspace/sl_report/configs/config.yaml`
- Seed: 7641
- Sample size: 20000
- Train: 16000, Test: 3999
- R version: 4.5.0
- Knitted: 2026-05-23 17:01:50